Introduction to Applied Cryptography – LDACS1310
Exercises Session 4

Questions

Question 1: Existentially UnForgeable under an adaptive Chosen-Message Attack (EUF-CMA)
Let $\mathsf{MAC} = (\mathsf{Gen}, \mathsf{Mac}, \mathsf{Vrfy})$ be a variable-length MAC that is existentially unforgeable under an adaptive chosen-message attack. Consider the following schemes $\mathsf{MAC}' = (\mathsf{Gen}', \mathsf{Mac}', \mathsf{Vrfy}')$ based on MAC with $\mathsf{Gen}' = \mathsf{Gen}$ and where Mac' are defined as follows:

a. $\mathsf{Mac}'_k(m) := (\mathsf{Mac}_k(m), \mathsf{Mac}_k(m \oplus 0...01))$

b. $\mathsf{Mac}'_k(m) := \mathsf{Mac}_k(\overset{l}{\underset{i=1}{\oplus}} m_i)$

c. $\mathsf{Mac}'_k(m) := (\mathsf{Mac}_k(m_1), ..., \mathsf{Mac}_k(m_l))$

d. $\mathsf{Mac}'_k(m) := (\mathsf{Mac}_k(m_1), \mathsf{Mac}_k(m_1||m_2), ..., \mathsf{Mac}_k(m_1||...||m_l))$

Show that those scheme are not existentially unforgeable under an adaptive chosen-message attack. (In some cases m is parsed in $m_1, ..., m_l$ with $|m_1| = ... = |m_{l-1}| = n$, $|m_l| \leq n$ and $m_1||...||m_l = m$ ($||$ is the concatenation) where n is the security parameter.)

Question 2: Unsuccessful MAC length extension
Let F be a PRF. Below, we describe three *insecure variable-length* message authentication codes, Π_1, Π_2 and Π_3, which all use the same key generation algorithm Gen. The message space is *any number* of message blocks in $\{0,1\}^n$, where n is the security parameter.

$\mathsf{Gen}(1^n)$ outputs a random key $k \leftarrow \{0,1\}^n$.

The scheme Π_3 is built from Π_2 which is itself built from Π_1 as an (unsuccessful) attempt to "patch" the previous scheme:

$\Pi_1 = (\mathsf{Gen}, \mathsf{Mac}^1, \mathsf{Vrfy}^1)$**:** *"Deterministic MAC – Chaining PRFs"*
$\mathsf{Mac}^1_k(m_1, \ldots, m_\ell)$ computes $t_1 = F_k(m_1)$ as well as $t_i = F_k(m_i \oplus t_{i-1})$, for $i = 2$ to ℓ, and returns $t := t_\ell$ (note that only the last block is returned).
$\mathsf{Vrfy}^1_k((m_1, \ldots, m_\ell), t)$ outputs 1 if $\mathsf{Mac}^1_k(m_1, \ldots, m_\ell) = t$, and 0 otherwise.

$\Pi_2 = (\mathsf{Gen}, \mathsf{Mac}^2, \mathsf{Vrfy}^2)$**:** *"Padding a random message block in the end"*
$\mathsf{Mac}^2_k(m_1, \ldots, m_\ell)$ first picks a random $r \leftarrow \{0,1\}^n$ and then runs $t = \mathsf{Mac}^1_k(m_1, \ldots, m_\ell, r)$ and outputs (r, t).
$\mathsf{Vrfy}^2_k((m_1, \ldots, m_\ell), (r, t))$ outputs 1 if $\mathsf{Mac}^1_k(m_1, \ldots, m_\ell, r) = t$, and 0 otherwise.

$\Pi_3 = (\mathsf{Gen}, \mathsf{Mac}^3, \mathsf{Vrfy}^3)$**:** *"Padding a random message block in the beginning"*
$\mathsf{Mac}^3_k(m_1, \ldots, m_\ell)$ first picks a random $s \leftarrow \{0,1\}^n$ and then runs $(r, t) = \mathsf{Mac}^2_k(s, m_1, \ldots, m_\ell)$ and outputs (r, s, t).
$\mathsf{Vrfy}^3_k((m_1, \ldots, m_\ell), (r, s, t))$ outputs 1 if $\mathsf{Mac}^1_k(s, m_1, \ldots, m_\ell, r) = t$, and 0 otherwise.

a) Describe $\mathsf{Mac}^3_k(m_1, \ldots, m_\ell)$ explicitly in terms of computations of F_k (and $\oplus$ of course).

b) Show the correctness of Π_3.

c) Mount a forgery attack on these MACs.

Question 3:

Let F be a pseudorandom function, G be a pseudorandom permutation, T be a public n-bit constant, k be a n-bit secret key, m be a n-bit message, IV be a n-bit random value chosen by the party computing the encryption (resp. MAC) before each operation. Among the following algorithms, determine the ones that would be acceptable in a corresponding scheme and justify your answer.

a) $\mathsf{Mac}_k(m) := F_k(m)$ as a MAC scheme existentially unforgeable under an adaptive chosen-message attack.

b) $\mathsf{Mac}_k(m) := F_k(m \oplus T)$ as a MAC scheme existentially unforgeable under an adaptive chosen-message attack.

c) $\mathsf{Mac}_k(m) := (IV, G_k(m \oplus IV \oplus T))$ as a MAC scheme existentially unforgeable under an adaptive chosen-message attack.